3) $q_1 = [1,2,3]$ $\quad k_1 = [1,1,1]$ $\quad v_1 = [2,0,1]$

$q_2 = [2,3,2]$ $\quad k_2 = [0,0,0]$ $\quad v_2 = [3,0,0]$

$q_3 = [5,6,7]$ $\quad k_3 = [2,2,0]$ $\quad v_3 = [1,2,2]$

$\sqrt{|k_1|} = 2$

a·i) $a_{ij} = q_i \cdot k_j$

$a_{11} = 1(1) + 2(1) + 3(1) = 1 + 2 + 3 = 6$

$a_{12} = 1(0) + 2(0) + 3(0) = 0 + 0 + \quad = 0$

$a_{13} = 1(2) + 2(2) + 3(0) = 2 + 4 + 0 = 6$

a·ii) $a_{11} = a_{11} / \sqrt{|k_1|} = 6/2 = 3$

$a_{12} = a_{12} / \sqrt{|k_1|} = 0/2 = 0$

$a_{13} = a_{13} / \sqrt{|k_1|} = 6/2 = 3$

$a_1 = [3 \ 0 \ 3]$

b) $\Sigma_k a_{ik} = a_{11} + a_{12} + a_{13} = 3 + 0 + 3 = 6$

$a_{11} = \frac{3}{6} = 0.5$

$a_{13} = \frac{3}{6} = 0.5$

$a_{12} = \frac{0}{6} = 0$

$a_1 = [0.5, 0, 0.5]$

c) $z_i = \sum_j a_{ij} v_j$

$a_{11} v_1 = 0.5[2, 0, 1] = [1, 0, 0.5]$

$a_{12} v_2 = 0[3, 0, 0] = [0, 0, 0]$

$a_{13} v_3 = 0.5[1, 2, 2] = [0.5, 1, 1]$

$z_1 = [1 + 0 + 0.5, 0 + 0 + 1, 0.5 + 0, + 1]$

$$\boxed{z_1 = [1.5, 1, 1.5]}$$